Midterm Q1-Q2

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Problem 1

The following code simulates and plots 5000 trajectories of Geometric Brownian motion with starting value $X_0 = 100$, growth rate$\mu = 0.10$, and volatility $\sigma = 0.21$.

```
import numpy as np
import matplotlib.pyplot as plt

n_paths = 5000
n_steps = 100
T = 1.0
dt = T / n_steps
Sqrtdt=np.sqrt(dt)
rng = np.random.default_rng() #initialize rand numb generator
X = np.zeros((n_paths, n_steps+1)) #Allocate a zero array of trajectories

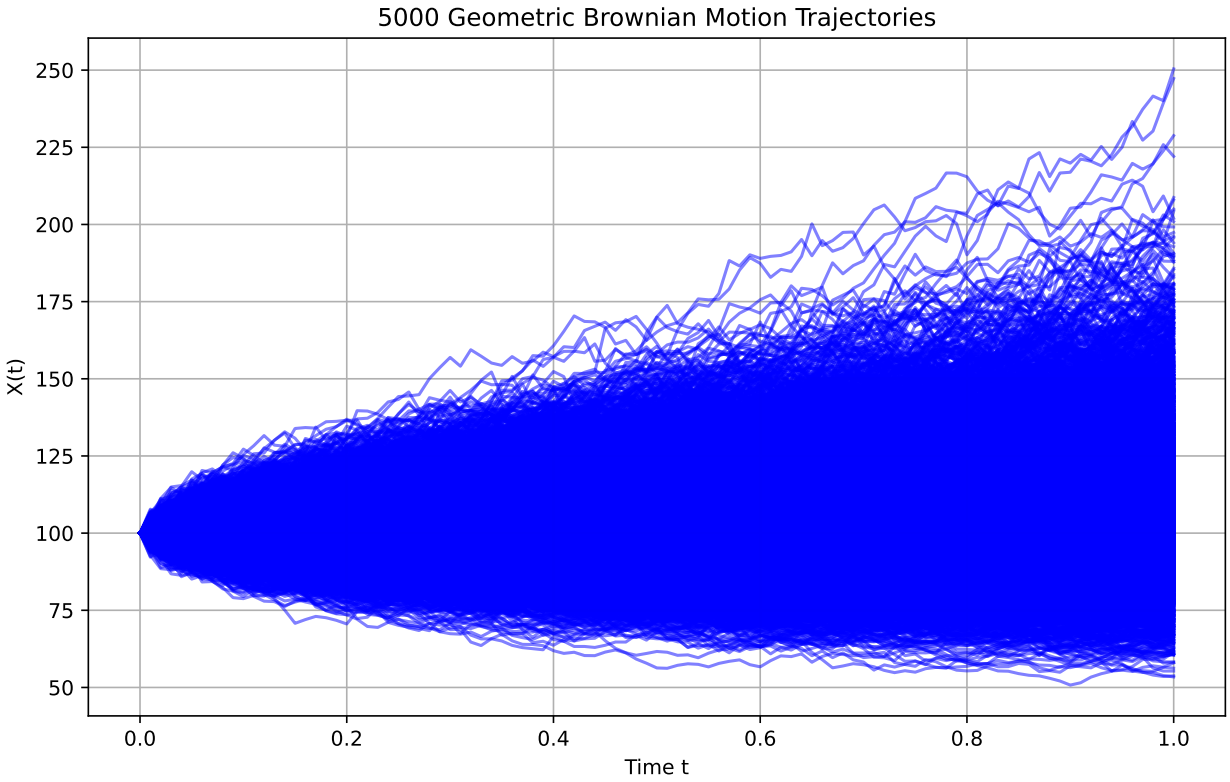
#GBM setup
X0 = 100
mu = 0.10
sigma = 0.21

X[:, 0] = X0 # initial starting point to X0
for i in range(n_paths):
    for j in range(1,n_steps+1):
      z = rng.normal(loc=0, scale=1)
      # GBM: X_(t + dt) = X_t * exp{(mu - 1/2 * sigma^2)dt + sigma * sqrt(dt)z}
      X[i, j] = X[i, j-1] * np.exp((mu - 0.5 * sigma**2) * dt + sigma * Sqrtdt * z)

# Time axis (n_steps+1) steps from 0 to T
t = np.linspace(0, T, n_steps + 1)

# Plotting
plt.figure(figsize=(10, 6))
plt.plot(t, X.T, alpha=0.5, color='blue')
plt.title('5000 Geometric Brownian Motion Trajectories')
plt.xlabel('Time t')
plt.ylabel('X(t)')
plt.grid(True)
```

```
plt.show()
```



Problem 2